⊛. KNN CLASIFIER Working principle.

⊛. What is KNN?

KNN stands for K-Nearest Neighbors

→ KNN is a supervised machine learning Algorithm.

→ KNN is used for both classification & Regression.

⊛. KNN- classifier Working principle

→ First KNN takes a snapshot of complete training data.

→ when a new test data point comes it calculates the distance between new data point and all the training data.

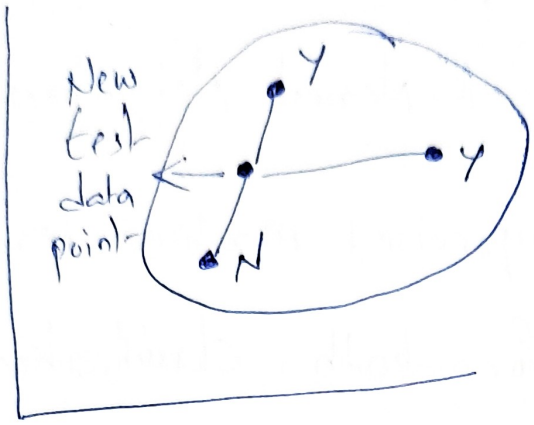
→ Now based on the K-value which user gives it picks the nearest K neighbors.

→ Now KNN classifies the new data points in Two methods.

method 1:- UNIFORM method / Normal.

method 2:- distance method / weighted.

→ Normal / uniform method :-



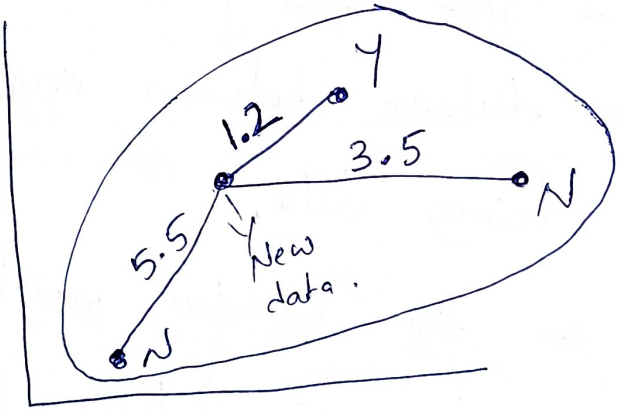
K = 3

distance = manhattan

2 - Yes | 1 - No

majority is Yes so KNN will classify the test data point as Y.

→ Weighted / distance method :- (Nearest data points are given more importance).



T.V	distance	weighted 1/d
Y	1.2	$\frac{1}{1.2} = 0.83$
N	3.5	$\frac{1}{3.5} = 0.28$
N	5.5	$\frac{1}{5.5} = 0.18$

Final

Yes weight :- 0.83

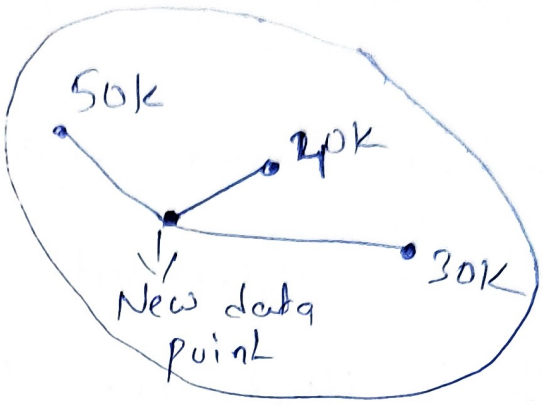
No weight :- 0.28 + 0.18

=> 0.46

since Yes has more weight knn classifies the data point as **Yes**

KNN Regressor:-

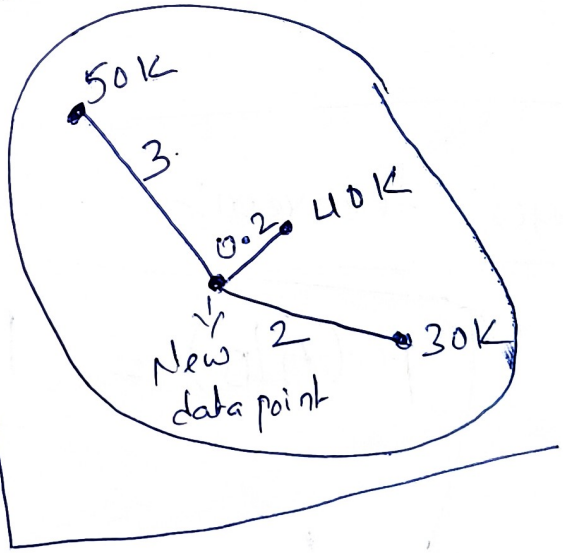
-> Normal / uniform



Here the model takes the average value of the nearest neighbors to predict the test data point value.

Final_predict_value → $\frac{50K+40K+30K}{3}$ => 40K

-> Weighted / distance method:-



final prediction will be given by formula.

$$\frac{\Sigma w_i \times y_i}{\Sigma w_i}$$

T.V	distance	weight (Yd) distance
40K	0.2	$\frac{1}{0.2} = 5$
30K	2	$\frac{1}{2} = 0.5$
50K	3	$\frac{1}{3} = 0.3$

Final prediction

$$\frac{5 \times 40K + 0.5 \times 30K + 0.3 \times 50K}{5 + 0.5 + 0.3}$$

$$= 39655$$

Naive Bayes Algorithm:-

-> Naives Bayes algorithm uses Bayes theorem
-> It's a supervised learning Algorithm.
-> It is used for classification.
-> It has strong Assumptions.
1. Data ~~the~~ is normal.
2. All the Independent Variables are Independent from each other.

Bayes theorem:-

$$P(A/B) = \frac{P(B/A)\ P(A)}{P(B)}$$

Derivation:-

1. Conditional probability:-

$$P(A/B) = \frac{P(A \cap B)}{P(B)} \quad \text{---} (1)$$

2. write the same for reverse.

$$P(B/A) = \frac{P(B \cap A)}{P(A)}$$

$(P(A \cap B) = P(B \cap A).$

$$P(B/A) = \frac{P(A \cap B)}{P(A)} \quad -(2)$$

(since $P(A \cap B)$ & $P(B \cap A)$ are same we write $P(B \cap A)$ as $P(A \cap B)$)

Rearranging eq-(1)

$$P(A \cap B) = P(A/B) \cdot P(B) \quad -(3)$$

Rearranging eq-(2)

$$P(A \cap B) = P(B/A) \cdot P(A) \quad -(4)$$

Now equate eq-(3) & eq-(4)

$$P(A/B) \cdot P(B) = P(B/A) \cdot P(A)$$

* Final Derivation:-

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

Consider Independent Variable Temparature & Target Variable play yes / play No.

Given the Temparature predict whether children play or not.

Temp	play.
High	no
low	yes
med	no
med	yes
low	yes
low	yes
high	no
low	no
med	no
med	yes
low →	? (predict)

contingency table.

	no(0)	yes(1)	
high	2^{high}	0^{high}	2 n(high)
low	1^{low}	3^{low}	4 n(low)
med	2^{med}	2^{med}	4 n(med)
	5	5	10
	n(no)	n(yes)	

Now let's write the equations.

$$prob\left(\frac{yes}{temp=low}\right) :- \frac{P(low/yes)\,P(yes)}{P(low)}$$

$$prob\left(\frac{yes}{temp=med}\right) :- \frac{P(med/yes)\,P(yes)}{P(med)}$$

$$prob\left(\frac{yes}{temp=High}\right) :- \frac{P(High/yes)\,P(yes)}{P(High)}$$

let's consider. Given temp = low predict play.

$$P(\text{Yes}/\text{low}) :- \frac{P(\text{low}/\text{Yes})\,P(\text{Yes})}{P(\text{low})}$$

$$P(\text{low}/\text{Yes}) :- \frac{P(\text{low} \cap \text{Yes})}{n(\text{Yes})} \Rightarrow \frac{n(\text{low} \& \text{Yes})}{n(\text{Yes})}$$

$$:- \frac{n(\text{low} \& \text{Yes})}{n(\text{Yes})} \rightarrow \frac{3}{5}$$

$$P(\text{Yes}) :- \frac{n(\text{Yes})}{\text{Total}} = \frac{5}{10}$$

$$P(\text{low}) :- \frac{n(\text{low})}{\text{Total}} = \frac{4}{10}$$

Substitute every thing.

$$P(\text{Yes}/\text{low}) :- \frac{\frac{3}{5} \times \frac{5}{10}}{4/10} = \frac{3/10}{4/10}$$

$$\frac{3}{10} \times \frac{10}{4} \Rightarrow \frac{3}{4} = 0.75$$

Now Apply Threshold & classify the data point.

(*) Consider Numerical Variable (INDEPENDENT).

Temp	play
30	N
33	N
32	N
38	N
27	Y
26	Y
25	Y
29	Y
31 →	?

$P(yes) = \frac{4}{8}$

$P(No) = 4/8.$

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

$$P(yes/31) = \frac{P(31/yes) \times P(yes)}{P(31)}$$

$$P(yes/31) = \frac{P(31/yes) \times P(yes)}{P(31/yes)\,P(yes) + P(31/No)\,P(No)}$$

-) Now we need to calculated

$P(31/yes) = ?$

$P(yes) = 4/8$

$P(31/No) = ?$

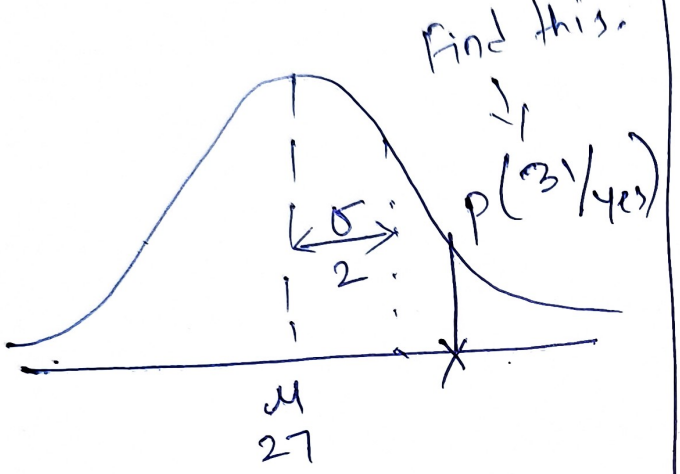
$P(No) = 4/8.$

} we will use Normal distribution to find $P(31/yes)$ & $P(31/No)$.

-) lets separate the data.

Temp_Yes

[27, 26, 25, 29]



Now Assume

$\mu = 27$

$\sigma = 2$

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$p(31/yes) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{31-27}{2}\right)^2}$$

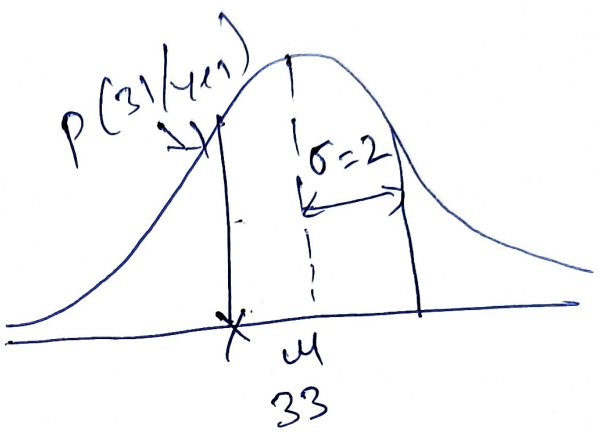
$$p(31/yes) = \frac{1}{8.8} e^{-\frac{1}{2}(4)}$$

$\Rightarrow$ 0.11

Assume

Temp_No

[30, 33, 32, 38]



Now Assume.

$\mu = 33$ $\sigma = 2$

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(x-\mu/\sigma)^2}$$

$$p(31/yes) \Rightarrow \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{31-33}{2}\right)^2}$$

Assume

$p(31/yes) = 0.34$

Now Highest prob value will be considered.